



## **Experiment 6**

### **AIM: Set Up NGINX Load Balancing**

Step 1: Download from <https://git-scm.com>

Step 2: C:\Users\Admin>\$ node -v

v22.13.1

Step 3: Create a Project

C:\Users\Admin>mkdir load-balance-demo

C:\Users\Admin>cd load-balance-demo

C:\Users\Admin\load-balance-demo>npm init -y

Wrote to D:\load-balance-demo\package.json:

```
{
  "name": "load-balance-demo",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" fifi exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
```

C:\Users\Admin\load-balance-demo>npm install express added 65 packages,

and audited 66 packages in 3s

22 packages are looking for funding

run `npm fund` for details

found 0 vulnerabilities

Step 4: Create server.js

Server.js for server1

```
const express = require("express");  
const app = express();  
const PORT = process.env.PORT || 3000;  
app.get("/", (req, res) => {  
  res.send(`Response from Server running on port ${PORT}`);  
});  
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

Server.js for server2

```
const express = require("express");  
const app = express();  
const PORT = process.env.PORT || 3001;  
app.get("/", (req, res) => {  
  res.send(`Response from Server running on port ${PORT}`);  
});  
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

Run the application

node server.js

Create a Folder `load Balancing`

Add `node.js` project

change `index.js` file (server - 1)

port : 3000

Add `node.js` project

change `index.js` file (server - 2)

port : 3001

Run Multiply Servers

Terminal 1:

At `PORT=3000`

`node server.js`

Terminal 2:

at `PORT=3001`

`node server.js`

So we have now:

`http://localhost:3000`

`http://localhost:3001`

Step 5: Configure `nginx` for load balancing

```
http {
```

```
    upstream myapp
```

```
{ server
```

```
    localhost:3000; server
```

```
    localhost:3001;
```

```
}
```

```
server {
```

```
listen 8081;

server_name localhost;

location / {
    proxy_pass http://myapp;
}

}
```

Restart Nginx

```
nginx -s stop
```

Nginx

Now open:

<http://localhost:8081>

Refresh multiple times

You can check:

Response from Server running on port 3000

Response from Server running on port 3001

Make Server 1 slow

Modify Server running on PORT 3000

```
app.get("/", (req, res) =>
{ setTimeout(() => {
    res.send("Slow Server - Port 3000");
}, 5000);
});
```

So Every request will wait 5 seconds before responding.

Keep Server 2 normal

```
app.get("/", (req, res) => {  
  res.send("Fast Server - Port 3001");  
});
```

Do Testing

Open: <http://localhost:8080>

Now refresh quickly multiple times.

Even though server 3000 is slow,

NGINX still keeps sending every alternate request to it.

Step 5: Now Change default algo of nginx (i.e. from round robin) to least Connections and test again

Modify nginx.conf file:

```
upstream myapp  
{ least_conn;  
  server localhost:3000;  
  server localhost:3001;  
}
```

Restart NGINX: (use below commands)

```
nginx -s stop
```

```
nginx
```

Open: <http://localhost:8080>

Refresh quickly multiple times quickly.

When slow server (3000) is busy for 5 seconds:

It has an active connection.

So NGINX sees:

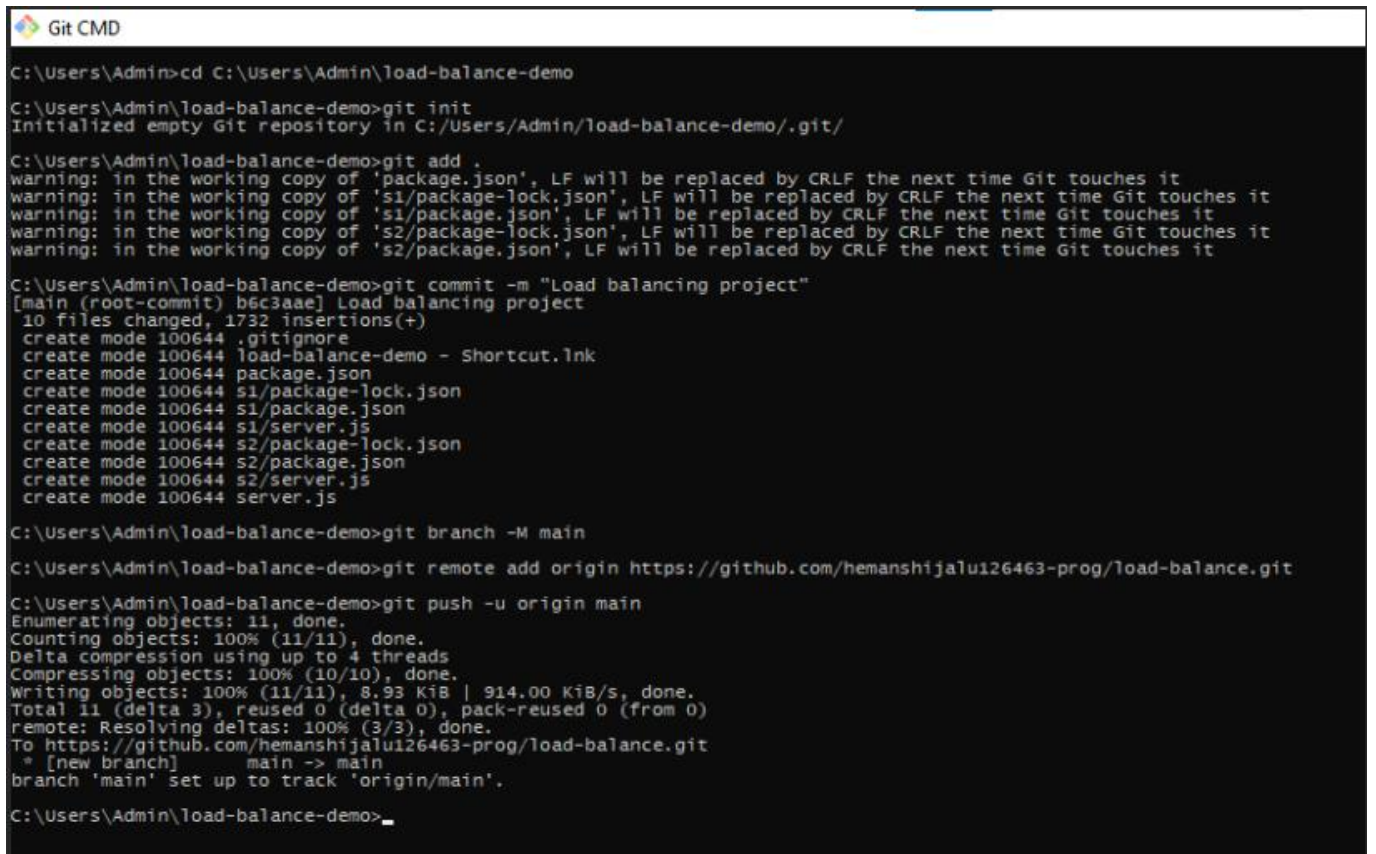
Server 3000 → 1 active connection

Server 3001 → 0 active connections

So it sends new requests to:

Fast Server (3001)

Output:



```
C:\Users\Admin>cd C:\Users\Admin\load-balance-demo
C:\Users\Admin\load-balance-demo>git init
Initialized empty Git repository in C:/Users/Admin/load-balance-demo/.git/

C:\Users\Admin\load-balance-demo>git add .
warning: in the working copy of 'package.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 's1/package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 's1/package.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 's2/package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 's2/package.json', LF will be replaced by CRLF the next time Git touches it

C:\Users\Admin\load-balance-demo>git commit -m "Load balancing project"
[main (root-commit) b6c3aae] Load balancing project
10 files changed, 1732 insertions(+)
create mode 100644 .gitignore
create mode 100644 load-balance-demo - Shortcut.lnk
create mode 100644 package.json
create mode 100644 s1/package-lock.json
create mode 100644 s1/package.json
create mode 100644 s1/server.js
create mode 100644 s2/package-lock.json
create mode 100644 s2/package.json
create mode 100644 s2/server.js
create mode 100644 server.js

C:\Users\Admin\load-balance-demo>git branch -M main
C:\Users\Admin\load-balance-demo>git remote add origin https://github.com/hemanshijalu126463-prog/load-balance.git
C:\Users\Admin\load-balance-demo>git push -u origin main
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 4 threads
Compressing objects: 100% (10/10), done.
Writing objects: 100% (11/11), 8.93 KiB | 914.00 KiB/s, done.
Total 11 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/hemanshijalu126463-prog/load-balance.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Users\Admin\load-balance-demo>
```

The screenshot shows a GitHub repository named 'load-balance' owned by 'hemanshijalu126463-prog'. The repository is public and has 1 branch and 0 tags. The main branch is selected. The repository contains a 'Load balancing project' with a commit hash 'b6c3aae' made 2 minutes ago, containing 1 commit. The file list includes 's1', 's2', '.gitignore', 'load-balance-demo - Shortcut.lnk', 'package.json', and 'server.js', all committed 2 minutes ago. A 'README' file is also present. The right sidebar shows the 'About' section with no description, website, or topics provided. It also shows 0 stars, 0 watching, and 0 forks. The 'Releases' section indicates no releases have been published, and the 'Packages' section indicates no packages have been published.